

GFG 160 - Complete DSA Question List

A topic-wise roadmap of 160 Data Structures and Algorithms problems to solve in Java, following the selected GeeksforGeeks order.

Total Questions	160
Total Topics	19
Language Plan	Java

No.	Topic	Question Range
1	Arrays	1-13
2	Strings	14-20
3	Sorting	21-27
4	Searching	28-35
5	Matrix	36-41
6	Hashing	42-49
7	Two Pointer Technique	50-60
8	Prefix Sum	61-65
9	Linked List	66-75
10	Backtracking	76-78
11	Trees and BST	79-93
12	Heap	94-97
13	Stack	98-106
14	Queue and Deque	107-108
15	Dynamic Programming	109-131
16	Greedy	132-135
17	Graph	136-152
18	Tries	153-155

No.	Topic	Question Range
19	Bit Manipulation	156-160

1. Arrays - Questions 1-13

1	Second Largest Element
2	Move All Zeros to End
3	Reverse an Array
4	Rotate an Array
5	Next Permutation
6	Majority Element II
7	Stock Buy and Sell - No Limit
8	Stock Buy and Sell - One Limit
9	Minimize the Heights
10	Maximum Sum Subarray
11	Maximum Product Subarray
12	Maximum Sum Circular Subarray
13	First Missing Positive

2. Strings - Questions 14-20

14	String to Integer - Implement atoi()
15	Add Two Binary Strings
16	Anagram Check
17	First Non-Repeating Character
18	KMP Algorithm for Pattern Searching
19	Minimum Characters to Add for Palindrome
20	String Rotations of Each Other

3. Sorting - Questions 21-27

21	Sort an Array of 0s, 1s and 2s
22	Count Inversions
23	Merge Overlapping Intervals
24	Insert and Merge Interval
25	Find Non-Overlapping Intervals
26	Merge Two Sorted Arrays with Constant Extra Space
27	Top K Frequent Elements

4. Searching - Questions 28-35

28	Count Occurrences in a Sorted Array
29	Minimum in a Sorted and Rotated Array
30	Search in a Sorted and Rotated Array
31	Peak Element
32	K-th Element of Two Merged Sorted Arrays
33	Aggressive Cows
34	Allocate Minimum Pages
35	K-th Missing Positive Number

5. Matrix - Questions 36-41

36	Spiral Traversal
37	Rotate Matrix
38	Search in a Sorted Matrix
39	Search in a Row-Wise Sorted Matrix
40	Search in a Row-Wise and Column-Wise Sorted Matrix
41	Set Matrix Zeroes

6. Hashing - Questions 42-49

42	Pair with Given Sum - Two Sum
43	Count Pairs with Given Sum
44	Intersection of Two Sets
45	Union of Two Sets
46	Longest Consecutive Subsequence
47	Group Anagrams
48	Count Subarrays with Given Sum
49	Count Subarrays with Given XOR

7. Two Pointer Technique - Questions 50-60

50	Check for Pair Sum in Sorted Array
51	Triplets in Sorted Array
52	Triplets in Unsorted Array
53	Print All Pairs with Given Sum
54	Closest Pair
55	Count Pairs in Sorted Array
56	Subarray with Given Sum
57	Count Distinct Elements in Every Window of Size K
58	Longest Substring with Distinct Characters
59	Trapping Rain Water
60	Container with Most Water

8. Prefix Sum - Questions 61-65

61	Equilibrium Point
62	Longest Subarray with Given Sum
63	Longest Subarray with Equal Number of 0s and 1s
64	Product of Array Except Self
65	Starting Petrol Pump for Circular Tour

9. Linked List - Questions 66-75

66	Reverse a Linked List
67	Rotate a Linked List
68	Merge Two Sorted Linked Lists
69	Reverse a Linked List in Groups
70	Sum Two Linked Lists
71	Clone a Linked List
72	Detect Cycle
73	First Node of the Cycle
74	Remove Cycle
75	LRU Cache

10. Backtracking - Questions 76-78

76	Permutations of a String
77	N-Queens
78	Solve the Sudoku

11. Trees and BST - Questions 79-93

79	Level Order Traversal
80	Height of a Binary Tree
81	Diameter of a Tree
82	Invert Binary Tree
83	Construct Tree from Inorder and Preorder
84	Morris Traversal
85	Boundary Traversal
86	Maximum Path Sum
87	Count All K-Sum Paths in Binary Tree
88	Check Whether Binary Tree Is a Valid BST
89	K-th Smallest Element in BST
90	Pair with Given Sum in BST
91	Correct a BST with Two Swapped Nodes
92	Lowest Common Ancestor in BST
93	Serialize and Deserialize a Binary Tree

12. Heap - Questions 94-97

94	K Largest Elements
95	K Closest Points to Origin
96	Merge K Sorted Lists
97	Median of a Stream

13. Stack - Questions 98-106

98	Valid Parentheses
99	Longest Valid Parentheses
100	Next Greater Element
101	Stock Span Problem
102	Largest Area in a Histogram
103	Maximum and Minimum of Every Window Size
104	Min Stack
105	Evaluate Reverse Polish Notation
106	Decode the String

14. Queue and Deque - Questions 107-108

107	Sliding Window Maximum
108	Maximum Score with Jumps of at Most Length K

15. Dynamic Programming - Questions 109-131

109	Longest Increasing Subsequence
110	Longest String Chain
111	Longest Common Subsequence
112	Longest Palindromic Subsequence
113	Longest Palindromic Substring
114	Count Palindromic Substrings
115	Edit Distance
116	Stair Climbing
117	Weighted Stair Climbing
118	0-1 Knapsack
119	Coin Change - Count Ways
120	Coin Change - Minimum Coins
121	Minimum Jumps to Reach End
122	Subset Sum Problem
123	Partition Equal Subset Sum
124	Stock Buy and Sell - Limit of K Transactions
125	Stock Buy and Sell - Limit of Two Transactions
126	House Robbing
127	House Robbing - Circular
128	Count Possible Decodings of a Digit Sequence
129	Matrix Chain Multiplication
130	Boolean Parenthesization
131	Word Break

16. Greedy - Questions 132-135

132	Minimum Platforms
133	Weighted Job Scheduling
134	Job Sequencing
135	Maximize Partitions with Unique Characters

17. Graph - Questions 136-152

136	Depth First Search - DFS
137	Breadth First Search - BFS
138	Minimum Time to Rot Oranges
139	Cycle in an Undirected Graph
140	Islands in a Graph
141	Topological Sorting
142	Cycle in a Directed Graph
143	Bridges in a Graph
144	Articulation Points
145	Minimum Cost to Connect All Points
146	Dijkstra's Algorithm
147	Flood Fill
148	Clone a Graph
149	Alien Dictionary
150	Distance from Source with Negative Weight Cycle
151	City with the Smallest Number of Neighbours at a Threshold Distance
152	Email Account Merging

18. Tries - Questions 153-155

153	Implement Trie
154	Word Search
155	Maximum XOR of Two Numbers in an Array

19. Bit Manipulation - Questions 156-160

156	Find the Only Repetitive Element Between 1 and N-1
157	Find the Missing Number
158	Unique Number I
159	Unique Number II
160	Unique Number III

Note: Each Java solution should use a unique file name and matching public class name to avoid Main.java or duplicate-class errors.